NETWORKED CONTROL

Marcello Farina

PART 1: BASICS

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A. CONTINUOUS-TIME SYSTEMS

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$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \qquad u \in \mathbb{R}^m, \quad x \in \mathbb{R}^n, \quad y \in \mathbb{R}^p$$

a) LAGRANGE EQUATION: GIVEN $x(t_0) = x_0$, $u(t)$

$$x(t) = \underbrace{e^{A(t-t_0)} x_0}_{\text{FREE MOTION}} + \underbrace{\int_{t_0}^{t} e^{A(t-\tau)} B u(\tau)\, d\tau}_{\text{FORCED MOTION}}$$

$$y(t) = \underbrace{C e^{A(t-t_0)} x_0}_{\text{FREE}} + \underbrace{C \int_{t_0}^{t} e^{A(t-\tau)} B u(\tau)\, d\tau + D u(t)}_{\text{FORCED}}$$

b) EQUILIBRIA: GIVEN $u(t) = \bar{u}$ $\forall t$

$$x(t) = \bar{x} \qquad \dot{x}(t) = 0 = \boxed{A\bar{x} + B\bar{u} = 0}$$

$$A\bar{x} = -B\bar{u}$$

IF A is invertible: $\bar{x} = -A^{-1}B\bar{u}$

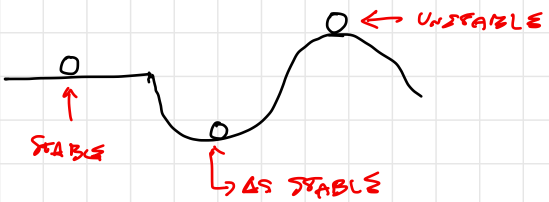
OTHERWISE NO OR ∞ SOLUTIONS

Note that A is invertible if
$\det(A) \neq 0$

Recall that
$\det(A) = \prod_{i=1}^{n} \lambda_i$ ⇝ EIGENVALUES

$\Rightarrow$ A IS INVERTIBLE IFF $\nexists\ \lambda_i = 0$

c) STABILITY



FOR LTI SYSTEMS STABILITY IS A STRUCTURAL (SYSTEM) PROPERTY $\Rightarrow$ PROPERTY OF A!

- <u>Theorem</u>

THE SYSTEM IS
- ASYMPTOTICALLY STABLE IFF $\mathrm{Re}(\lambda_i) < 0 \ \forall$ EIGENVALUES
  (A IS HURWITZ STABLE)
- UNSTABLE IF $\exists \lambda_i$ WITH $\mathrm{Re}(\lambda_i) > 0$

SOME USEFUL DEFINITIONS:

- CONSIDER A MATRIX $P = P^T$ (symmetric)

- $\underline{P > 0}$ POSITIVE DEFINITE IFF $\forall x \neq 0 \quad x^T P x > 0$

<u>EX 1</u> $P = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow x^T P x = x_1^2 + x_2^2 > 0$

<u>EX 2</u> $P = \begin{bmatrix} 1 & -\frac{1}{4} \\ -\frac{1}{4} & 1 \end{bmatrix} \Rightarrow x^T P x = [x_1 \ \ x_2] \begin{bmatrix} 1 & -\frac{1}{4} \\ -\frac{1}{4} & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$= \left[x_1 - \frac{1}{4} x_2 \quad x_2 - \frac{1}{4} x_1\right] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= x_1^2 + x_2^2 - \frac{1}{2} x_1 x_2 > 0 \quad (*)$$

IN FACT:

$$(x_1 - x_2)^2 = x_1^2 + x_2^2 - 2x_1x_2 \geq 0 \Rightarrow -x_1x_2 \geq -\frac{x_1^2}{2} - \frac{x_2^2}{2}$$

$$\Rightarrow (*) \ x_1^2 + x_2^2 - \frac{1}{2} x_1 x_2 \geq x_1^2 + x_2^2 + \frac{1}{2}\left(-\frac{x_1^2}{2} - \frac{x_2^2}{2}\right) = \frac{3}{4}\left(x_1^2 + x_2^2\right) > 0$$

- $\underline{P \geq 0}$ POSITIVE SEMIDEFINITE IFF $\forall x \neq 0 \quad x^T P x \geq 0$

<u>EX 3</u> $P = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow x^T P x = x_1^2 + x_2^2 - 2x_1x_2 = (x_1 - x_2)^2 \geq 0$!

- $P < 0$ NEGATIVE DEFINITE IFF $-P > 0$
- $P \leq 0$ NEGATIVE SEMIDEFINITE IFF $-P \geq 0$

- IF $P>0$ IS SYMMETRIC $\Rightarrow$ EIGENVALUES OF P ARE REAL AND POSITIVE

$$0 < \lambda_{min} < \dots < \lambda_{max}$$

ALSO:

$$\lambda_{min} \|x\|^2 \leq x^T P x \leq \lambda_{max} \|x\|^2 \qquad \|x\|^2 = x^T x$$

$$\hookrightarrow \|P\|_2 = \max_{x \in \mathbb{R}^n} \frac{\|Px\|}{\|x\|}$$

- Theorem (Lyapunov inequality)

A is Hurwitz stable iff $\exists\ P = P^T > 0$ such that $A^T P + PA < 0$

SKETCH OF THE PROOF

$\Rightarrow$ A Hurwitz $\Rightarrow$ $\exists\ P = P^T > 0$ s.t. $A^T P + PA < 0$ (*)

To prove it we take $Q = Q^T > 0$ arbitrary and show that

$$P = \int_0^{+\infty} e^{A^T t} Q e^{At} dt$$ fulfills the Lyapunov inequality (*)

$\hookrightarrow$ It is well defined since modes are $e^{\lambda_i t} \longrightarrow 0$ exponentially (in fact A is Hurwitz stable)

$$A^T P + PA = \int_0^{+\infty} \underbrace{\left(A^T e^{A^T t} Q e^{At} + e^{A^T t} Q e^{At} A\right)}_{\frac{d\left(e^{A^T t} Q e^{At}\right)}{dt}} dt$$

$$= \int_0^{+\infty} \frac{d}{\cancel{dt}} \left(e^{A^T t} Q e^{At}\right) \cancel{dt} = \int_0^{+\infty} d\left(e^{A^T t} Q e^{At}\right) = \left[e^{A^T t} Q e^{At}\right]_0^{+\infty} = 0 - Q$$

$$\Rightarrow A^T P + PA = -Q < 0 \quad !$$

($\Leftarrow$) $\exists\ P = P^T > 0$ s.t. $A^T P + PA < 0 \quad (*)$ $\Rightarrow$ A Hurwitz stable

In fact, in view of $(*)$: $v^T (A^T P + PA) v < 0 \quad (**) \quad \forall v \neq 0$

If v is an eigenvector of A corresponding with eigenvalue λ:

- $Av = \lambda v$
- $v^T A^T = (Av)^T = (\lambda v)^T = \lambda^* v^T$

complex conjugate of λ

REMARK: $(\ )^T$ is the <u>Hermite transposition</u>

- TRANSPOSITION +
- COMPLEX CONJUGATE OF COMPLEX ELEMENTS

From $(**)$: $v^T (A^T P + PA) v = v^T A^T P v + v^T P A v = \lambda^* v^T P v + \lambda v^T P v$

$$\overset{!}{=} (\lambda^* + \lambda) \underbrace{v^T P v}_{>0 \text{ since } P > 0} < 0$$

$\Downarrow$

THIS IS TRUE ONLY IF $\lambda^* + \lambda = 2\,\mathrm{Re}(\lambda) < 0$!

- NOTE THAT $V(x) = x^T P x$ IS A LYAPUNOV FUNCTION FOR $\dot{x} = Ax$

IN FACT

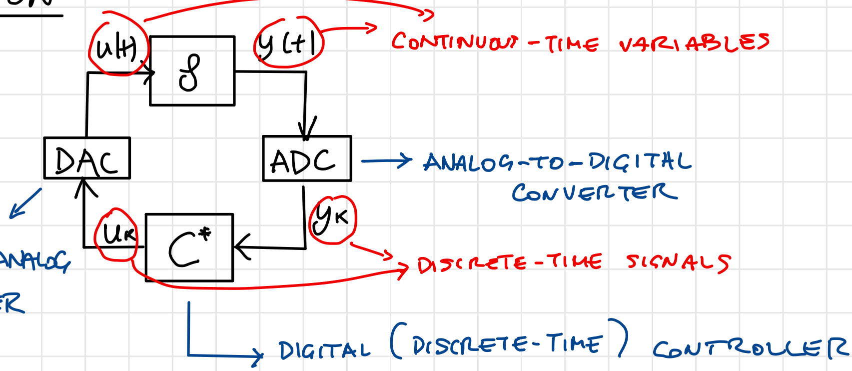
- $\lambda_{min} \|x\|^2 \leq x^T P x \leq \lambda_{max} \|x\|^2$
- $\dot{V}(x) = \dfrac{d}{dt}(x^T P x) = \dot{x}^T P x + x^T P \dot{x} = x^T A^T P x + x^T P A x$

$$= x^T (A^T P + PA) x < 0 \quad !$$

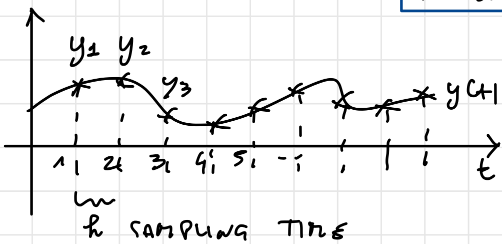
B. DISCRETE-TIME MODELS

a) DISCRETIZATION

A PROTOTYPICAL DIGITAL CONTROL SYSTEM IS →

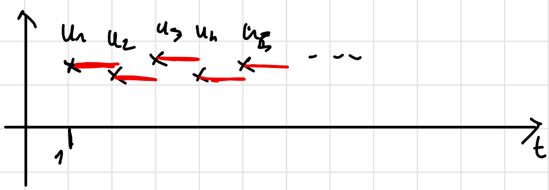


ADC - SAMPLING WITH $\boxed{h: \text{ SAMPLING TIME}}$



$$y_k = y(kh)$$

DAC (ZOH - ZERO ORDER HOLD)



$$u(t) \overset{(*)}{=} u_k \quad \text{FOR ALL } t \in [kh, (k+1)h)$$

DYNAMICAL EVOLUTION OF THE STATE VARIABLE $x(t)$ IN DISCRETE-TIME:

$$x_{k+1} = x((k+1)h) = e^{A((k+1)h - kh)} x(kh) + \int_{kh}^{(k+1)h} e^{A((k+1)h-\tau)} B u(\tau) d\tau$$

↳ THANKS TO THE LAGRANGE EQUATION

$$= e^{Ah} x_k + \underbrace{\int_{kh}^{(k+1)h} e^{A((k+1)h-\tau)} B \underbrace{u(\tau)}_{\to u_k \ (*)} d\tau}$$

$$\int_{kh}^{(k+1)h} e^{A((k+1)h-\tau)} d\tau \, B u_k = \overbrace{\int_0^h e^{A\tau} d\tau}^{\Gamma(h)} B u_k$$

$$\Rightarrow \quad x_{k+1} = \underbrace{e^{Ah}}_{F} x_k + \underbrace{\Gamma(\tau) B}_{G} u_k \qquad \text{(STATE EQUATION)}$$

OUTPUT EQUATION:

$$y(kh) = C x(kh) + D u(kh)$$

$$\downarrow$$

$$y_k = \underbrace{C}_{H} x_k + \underbrace{D}_{L} u_k$$

$$y_k = H x_k + L u_k$$

b) SOLUTION IN DISCRETE-TIME

$x_0 = x(t_0)$, u_k INPUT SEQUENCE

$$x_k = F^k x_0 + \sum_{i=1}^{k} F^{i-1} G u_{k-i}$$

$$= F^k x_0 + \sum_{i=1}^{k} F^{k-i} G u_{i-1}$$

$\downarrow$

FREE MOTION

$$F^k x_0 = \left(e^{Ah}\right)^k x_0 = e^{Akh} x_0$$

$$\downarrow$$

$$e^{At} x_0 \quad t = kh$$

c) EQUILIBRIA $\quad u_k = \bar{u} \; \forall k$

$\bar{x} \Rightarrow x_k = \bar{x} \; \forall k$

$$x_{k+1} = F x_k + G u_k$$

$$\bar{x} = F\bar{x} + G\bar{u} \Rightarrow (I-F)\bar{x} = G\bar{u}$$

$\bar{x}$ IS UNIQUE IFF $\underline{I-F}$ INVERTIBLE!

$F = e^{Ah}$ → ITS EIGENVALUES ARE $e^{\lambda_i h}$
$i = 1, \dots, n$

$I - F$ → ITS EIGENVALUES ARE $1 - e^{\lambda_i h}$

$I-F$ invertible when $1 - e^{\lambda_i h} \neq 0 \quad \forall \lambda_i$

$$\Downarrow$$

$$e^{\lambda_i h} \neq 1 \iff \lambda_i \neq 0 \quad \forall i$$

IF $\underline{I-F \text{ invertible}}$

$$\bar{x} = (I-F)^{-1} G \bar{u} \qquad G = \Gamma(h) B$$

$$\stackrel{!}{=} (I - e^{Ah})^{-1} \Gamma(h) \bar{u} \qquad \stackrel{!}{=} \int_0^h e^{A\tau} d\tau = A^{-1}(e^{Ah} - I)$$

$$\stackrel{!}{=} (I - e^{Ah})^{-1} (e^{Ah} - I) A^{-1} B \bar{u}$$

$$\stackrel{!}{=} -A^{-1} B \bar{u}$$

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$$\begin{cases} x_{k+1} = F x_k + G u_k \\ y_k = H x_k + L u_k \end{cases}$$

$F = e^{Ah}$ h: SAMPLING TIME

$G = \int_0^h e^{A\tau} d\tau B$

$H = C$

$L = D$

d) STABILITY (LTI)

- IN CONTINUOUS TIME: MODES $\Rightarrow e^{\lambda_i(A)t} \longrightarrow 0$ EXPONENTIALLY

$\dot{x} = Ax \quad (u=0) \quad \exists \mu > 0, \bar{\lambda} < 0$ S.T.

$\|x(t)\| \leq \mu e^{\bar{\lambda} t} \|x_0\| \Rightarrow$ FOR $t = kh$ (sampling times)

$\boxed{\|x_k\| = \|x(kh)\| \leq \mu (e^{\bar{\lambda} h})^k \|x_0\|}$

- IN DISCRETE-TIME: MODES $\Rightarrow \lambda_i(F)^k \longrightarrow 0$ EXPONENTIALLY

$x_{k+1} = F x_k \quad (u_k = 0) \quad \exists \mu > 0, \sigma \in [0,1)$ S.T.

$\boxed{\|x_k\| \leq \mu \sigma^k \|x_0\|}$

$\sigma = e^{\bar{\lambda} h}$

- Theorem

The discrete-time system is as. stable iff

$|\lambda_i(F)| < 1$ [F is Schur stable]

- Since $F = e^{Ah}$, the eigenvalues of F are $\lambda_i(F) = e^{\lambda_i(A) h}$. Therefore

$|\lambda_i(F)| < 1 \iff \mathrm{Re}(\lambda_i(A)) < 0$

[Schur stability of F] [Hurwitz stability of A]

- **Theorem** (Lyapunov inequality)

F is Schur stable $\iff \exists\, P^T = P > 0$ s.t. $F^T P F - P < 0 \quad (i)$

SKETCH OF THE PROOF

($\Rightarrow$) F is SCHUR STABLE $\Rightarrow \exists\, P = P^T > 0$ S.T. (i)

RECALL: IN CONTINUOUS-TIME

$$P = \int_0^{+\infty} e^{A^T\tau} Q\, e^{A\tau}\, d\tau \qquad Q^T = Q > 0$$

IN DISCRETE-TIME $\quad P = \sum_{k=0}^{+\infty} (F^T)^k Q F^k$

$\exists$ AND WELL-DEFINED BECAUSE F IS SCHUR STABLE

$$F^T P F = F^T \sum_{k=0}^{+\infty} (F^T)^k Q F^k F = \sum_{k=0}^{+\infty} (F^T)^{k+1} Q F^{k+1}$$

$$= \sum_{k=1}^{+\infty} (F^T)^k Q F^k + F^{T0} Q F^0 - Q = \underbrace{\sum_{k=0}^{+\infty} (F^T)^k Q F^k}_{=P} - Q = P - Q$$

$(F^0 = I)$

THEREFORE $\Rightarrow \quad F^T P F = P - Q$

$$\Rightarrow \boxed{F^T P F - P = -Q < 0}$$

($\Leftarrow$) $\exists\, P = P^T > 0$ S.T. $F^T P F - P < 0 \Rightarrow F$ is Schur stable

$$v^T (F^T P F - P) v < 0 \qquad \forall\, v \neq 0$$

THIS ALSO HOLDS IF v EIGENVECTOR OF F CORRESPONDING WITH EIGENVALUE λ

Note that $Fv = \lambda v$

$$v^T F^T = (Fv)^T = \lambda^* v^T \quad \text{(Hermite transpose)}$$

↳ COMPLEX CONJUGATE

$\boxed{\text{EXAMPLE}}$

$P = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} > 0 \qquad \forall v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{C} \qquad v^T P v > 0$

- IF $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\qquad v^T P v = [1 \ 1] \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 + 2 = 3 > 0$

- IF $v = \begin{bmatrix} 1 \\ j \end{bmatrix}$ $\qquad \cancel{v^T P v = [1 \ j] \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ j \end{bmatrix} = 1 + 2j^2 = 1 - 2 = -1 < 0}$

$\qquad v^T P v = [1 \ -j] \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ j \end{bmatrix} = 1 - 2j^2 = 1 + 2 = 3 > 0$

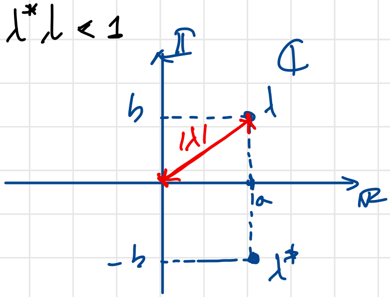
WE COMPUTE

$$\begin{aligned} v^T(F^T P F - P)v &= (v^T F^T) P (Fv) - v^T P v \\ &= \lambda^* v^T P v \lambda - v^T P v \\ &= (\lambda^* \lambda) v^T P v - v^T P v \\ &= [(\lambda^* \lambda) - 1] \underbrace{v^T P v}_{>0} < 0 \end{aligned}$$

$$\Rightarrow \lambda^* \lambda - 1 < 0 \ ! \quad \Rightarrow |\lambda|^2 = \lambda^* \lambda < 1$$

$\boxed{\text{OSS}}$ IN FACT, IF $\lambda = a + jb$

$\lambda^* = a - jb$



$$\lambda \lambda^* = (a + jb)(a - jb) = a^2 + b^2 = |\lambda|^2$$

To conclude, to check the stability of F:

- → Check if $\exists\, P = P^T > 0$ s.t. $F^T P F - P < 0$ (Lyapunov inequality)
- → Check if $|\lambda_i| < 1$ for all eigenvalues

↓

This can be checked directly considering the coefficients of the characteristic polynomial:

$$p_F(\lambda) = \det(\lambda I - F) = \varphi_0 \lambda^n + \varphi_1 \lambda^{n-1} + \dots + \varphi_n$$

To do so, we rely on the

<u>Jury criterion</u>

The Jury criterion consists of

1) Definition of the <u>Jury table</u>

$n+1$ rows

$$\begin{matrix} \varphi_0 & \varphi_1 & \dots & \varphi_n \\ \vdots & & & \end{matrix}$$

⇒ Triangular

Each row is computed based on the previous one using the following rule:

$$\begin{matrix} h_1 & h_2 & \dots & h_{v-1} & h_v \\ l_1 & l_2 & \dots & l_{v-1} & / \end{matrix} \qquad l_i = \frac{1}{h_1} \det \begin{bmatrix} h_1 & h_{v-i+1} \\ h_v & h_i \end{bmatrix}$$

If $h_1 = 0$ the table is not well defined

2) Application of the criterion:

F is Schur stable $\Longleftrightarrow$
- The Jury table is well-defined
- The elements of the first column of the Jury table are concordant (have the same sign)

EXAMPLE: CONSIDER $F \in \mathbb{R}^{2\times 2} \Rightarrow p_F(\lambda) = \lambda^2 + a\lambda + b$

1) JURY TABLE:

1	a	b
x_1	x_2	
y_1		

$$x_1 = \frac{1}{1}\det\begin{bmatrix} 1 & b \\ b & 1 \end{bmatrix} = 1 - b^2$$

$$x_2 = \frac{1}{1}\det\begin{bmatrix} 1 & a \\ b & a \end{bmatrix} = a(1-b)$$

$$y_1 = \frac{1}{x_1}\det\begin{bmatrix} x_1 & x_2 \\ x_2 & x_1 \end{bmatrix} = \frac{1}{1-b^2}\det\begin{bmatrix} 1-b^2 & a(1-b) \\ a(1-b) & 1-b^2 \end{bmatrix} = \frac{(1-b^2)^2 - a^2(1-b)^2}{1-b^2} = \frac{(1-b)^2\left[(1+b)^2 - a^2\right]}{1-b^2}$$

2) APPLICATION OF THE CRITERION

$$\begin{cases} 1 > 0 \\ x_1 = 1 - b^2 > 0 \quad (i) \\ y_1 = \dfrac{(1-b)^2}{1-b^2}\left((1+b)^2 - a^2\right) > 0 \quad (ii) \end{cases}$$

IS THE NECESSARY AND SUFFICIENT CONDITION FOR STABILITY OF F

> 0 IN VIEW OF (i)

(i): $\boxed{-1 < b < 1}$

(ii): $(1+b)^2 - a^2 < 0 \Rightarrow \boxed{-(1+b) < a < 1+b}$

$\Downarrow$

NECESSARY AND SUFFICIENT CONDITION FOR SCHUR STABILITY OF F

C. MATRIX INEQUALITIES

SO FAR WE HAVE INTRODUCED SOME MATRIX INEQUALITIES

$$\left.\begin{aligned} &P > 0 \\ &A^T P + PA < 0 \\ &F^T P F - P < 0 \end{aligned}\right\}$$ LINEAR (AFFINE, INDEED) MATRIX INEQUALITIES - LMI -

↳ They lead to convex problems (easy to solve)

THERE ARE 2 TYPES OF PROBLEMS INVOLVING LMIs:

1) LMI <u>FEASIBILITY PROBLEMS</u>:

$$\begin{cases} f_1(P) > 0 \\ \vdots \\ f_N(P) > 0 \end{cases}$$ (NOTHING TO MINIMIZE)

USED, E.G., FOR CHECKING STABILITY PROPERTIES

2) LMI <u>OPTIMIZATION PROBLEMS</u>:

$$\min_P \underbrace{\text{cost}(P)}_{\text{LINEAR WITH RESPECT TO } P} \quad \text{SUBJECT TO} \quad \begin{cases} f_1(P) > 0 \\ \vdots \\ f_N(P) > 0 \end{cases}$$

USED, E.G., FOR CONTROLLER DESIGN

EXERCIZES

1. $\dot{x} = -x + u$ $\qquad$ $\dot{x} = Ax + Bu$

a) $A = -1 = \lambda < 0 \Rightarrow$ AS. STABLE!

$\hookrightarrow$ MODE: $e^{\lambda t} = e^{-t}$

b) $t_0 = 0 \Rightarrow$ $\qquad$ $x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)} Bu(\tau)\,d\tau$

$x_{FR}(t) = e^{At}x_0 = e^{-t}$

Settling time $\simeq 5\tau$ $\qquad$ $\tau = \dfrac{1}{|Re(\lambda)|} = 1\ s$

$\stackrel{!}{=} 5\ s$

c) $A^T P + PA < 0$ $\qquad$ $P^T = P > 0$

$-P - P < 0 \qquad -2P < 0 \Rightarrow P > 0 \qquad P = 1$

Lyapunov function for $\boxed{\dot{x} = -x}$

$V(x) = x^T P x$ IS A LYAPUNOV FUNCTION

$\Rightarrow V(x) = x^2$

$\dot{V}(x) = 2x\dot{x} = 2x(-x) = -2x^2 < 0$

D) a) $u(t) = \bar{u} = 5 \Rightarrow$

$$x_{FO}(t) = \int_0^t e^{-(t-\tau)} 5\, d\tau = 5e^{-t}\int_0^t e^{\tau} d\tau = 5e^{-t}\left[e^{\tau}\right]_0^t$$

$$= 5e^{-t}\left(e^{t} - 1\right) = 5\left(1 - e^{-t}\right)$$

IN $t = kh$ $\quad x_{FO}(kh) = 5\left(1 - e^{-kh}\right)$

b) $u(t) = e^{-t} \rightarrow e^{\alpha t}$ $\quad \alpha = -1$ = EIGENVALUE

$\Rightarrow \quad u(t) = e^{\alpha t} \quad \Rightarrow \quad y(t) = G(\alpha) e^{\alpha t}$

$\alpha \neq \lambda$

G TR. FUNCT. OF SYSTEM

$\mathcal{L}(\dot{x}) = \mathcal{L}(x + u)$

$$s X(s) = -X(s) + U(s) \quad \Rightarrow \quad X(s) = \underbrace{\boxed{\frac{1}{s+1}}}_{G(s)} U(s)$$

$$X(s) = G(s) \frac{1}{s+1} = \frac{1}{(s+1)^2} \xrightarrow{\mathcal{L}^{-1}} x(t) = t e^{-t}$$

IN STATE-TRACE

$$x_{FO}(t) = \int_0^t e^{-(t-\tau)} e^{-\tau} d\tau = e^{-t} \int_0^t e^{\tau} e^{-\tau} d\tau$$

$$= e^{-t} \int_0^t d\tau = t e^{-t}$$

c) $u(t) = e^{-2t}$

$$x_{FO}(t) = \int_0^t e^{-(t-\tau)} e^{-2\tau} d\tau = e^{-t} \int_0^t e^{\tau} e^{-2\tau} d\tau$$

$$= e^{-t} \int_0^t e^{-\tau} d\tau$$

$$= e^{-t} (-1) [e^{-\tau}]_0^t$$

$$= -e^{-t} (e^{-t} - 1)$$

$$= e^{-t} - e^{-2t}$$

E) $\dot{x} = Ax + Bu \qquad u = Kx$

$$\downarrow \; Ax + BKx = (A+BK)x$$

$$A+BK = -1 + K < 0 \qquad \boxed{K < 1}$$

$$\tau = \frac{1}{|\mathrm{Re}(\lambda)|} = \frac{1}{|K-1|}$$

$$T_s = 5\tau \leq 0.5 \text{ s} \qquad \frac{5}{|K-1|} \leq 0.5$$

$$|K-1| \geq 10 \qquad K \leq -9$$

2. $\dot{x} = -x + u$ $\quad h$: SAMPLING TIME

$x_{k+1} = F x_k + G u_k$ $\quad F = e^{Ah}$

$G = \int_0^h e^{A\tau} d\tau \, B$

$F = e^{-h}$

$G = \int_0^h e^{-\tau} d\tau = -\left[e^{-\tau}\right]_0^h = -\left(e^{-h} - 1\right) = 1 - e^{-h}$

A) $\lambda_{DT} = \boxed{e^{-h} = F}$ $\quad |e^{-h}| < 1 \Rightarrow$ Schur stable

MODE : $\lambda_{DT}^k = (e^{-h})^k = e^{-kh}$

B) $x_k = \underbrace{F^k x_0}_{} + \sum_{i=1}^{k} F^{i-1} G u_{k-i+1}$

$x_{FR,k} = (e^{-h})^k = e^{-kh}$

C) $F^T P F - P < 0$

$e^{-h} P e^{-h} - P < 0$ $\quad e^{-2h} P - P < 0$

$\underbrace{(e^{-2h} - 1)}_{<0} P < 0$ $\quad P > 0$ ANY

$\hookrightarrow P = 1$

$V(x_k) = x_k^T P x_k$ $\quad x_{k+1} = \boxed{A x_k}$

$V(x_{k+1}) < V(x_k)$

$\hookrightarrow V(x_k) = x_k^2$

$\rightarrow < 1$

$V(x_{k+1}) = (e^{-h} x_k)^2 = e^{-2h} x_k^2 < x_k^2 = V(x_k)$

D) a) $u_k = \bar{u}$

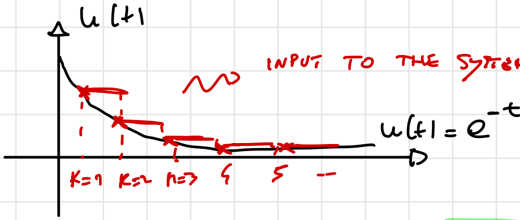
$$x_{F0,k} = \sum_{i=1}^{k} F^{i-1} G u_{k-i} = \sum_{i=1}^{k} (e^{-h})^{i-1}(1-e^{-h})5$$

$$= \underbrace{\sum_{i=0}^{k-1}(e^{-h})^{i}}_{\downarrow} (1-e^{-h})5$$

$$= \frac{1-e^{-kh}}{1-\cancel{e^{-h}}}\,(1-\cancel{e^{-h}})\cdot 5$$

$$= (1-e^{-kh})5$$

b) $u_k = e^{-kh}$



INPUT TO THE SYSTEM: PIECE-WISE CONSTANT SIGNAL, GENERATED BY THE SAMPLE-AND-HOLD ACTUATOR

$$x_k = \sum_{i=1}^{k}(e^{-h})^{i-1}(1-e^{-h})\,e^{-h(k-i)}$$

$$= (1-e^{-h})e^{-kh}\underbrace{\sum_{i=1}^{k}\cancel{(e^{-h})^{i-1}}\,\cancel{e^{-h(1-i)}}}_{\sum_{i=1}^{k} 1 = k}\, e^{h}$$

$$= (1-e^{-h})\,k\,e^{-(k-1)h}$$

c) SEE LECTURE NOTES

E) a) WHILE IN CONTINUOUS-TIME THE CONTROL LAW IS $u(t) = Kx(t)$, A DIGITAL CONTROLLER PROVIDES A NEW CONTROL ACTION AT EACH SAMPLING INSTANT, I.E., $u_k = Kx_k$

$$x_{k+1} = Fx_k + G u_k = (F + GK) x_k$$

$$x_{k+1} = \left(e^{-h} + (1 - e^{-h}) K\right) x_k$$

$$\left| e^{-h} + (1 - e^{-h}) K \right| < 1$$

$$-1 < e^{-h} + (1 - e^{-h}) K < 1$$

$$\begin{cases} e^{-h} + (1 - e^{-h}) K > -1 \\ e^{-h} + (1 - e^{-h}) K < 1 \rightsquigarrow (1 - \cancel{e^{-h}}) K < 1 - \cancel{e^{-h}} \end{cases}$$

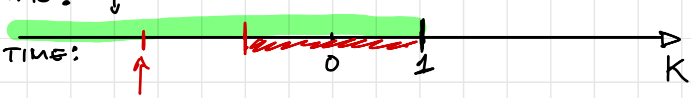
$$\Rightarrow \boxed{K < 1}$$

$$K(1 - e^{-h}) > -1 - e^{-h}$$

$$K > \frac{-1 - e^{-h} \pm e^{-h}}{1 - e^{-h}} = \underbrace{\boxed{\frac{-1 + e^{-h}}{1 - e^{-h}}}}_{-1} - \frac{2e^{-h}}{1 - e^{-h}} = -1 - \frac{2e^{-h}}{1 - e^{-h}}$$

CONT. TIME:

DISCRETE TIME:



$$-1 - \frac{2e^{-h}}{1 - e^{-h}} \xrightarrow{h \to +\infty} -1 \qquad -1 - \frac{2e^{-h}}{1 - e^{-h}} \xrightarrow{h \to 0} -\infty$$

b) $K = -9$? How small must h be to have

$F + GK$ <u>asymptotically stable</u>?

$$F + GK = e^{-h} + (1 - e^{-h})K$$
$$\stackrel{!}{=} e^{-h} - 9(1 - e^{-h})$$

$$\left| e^{-h} - 9(1 - e^{-h}) \right| < 1$$

- $e^{-h} - 9(1 - e^{-h}) < 1 \iff -9(1 - e^{-h}) < (1 - e^{-h})$

$\Rightarrow$ FOR ALL h!

- $e^{-h} - 9(1 - e^{-h}) > -1$

$$e^{-h}(1 + 9) > -1 + 9 = 8$$

$$e^{-h} > \frac{8}{10} \iff -h > \ln\left(\frac{8}{10}\right)$$

$$\Rightarrow \boxed{h < \left|\ln\left(\frac{8}{10}\right)\right|}$$

h must be sufficiency small!